```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Py
Works/Lab1.py"
Point 1: [4 8 5 9 7]
Point 2: [ 9 10 12 11 13]

Euclidean Distance: 10.863
Manhattan Distance: 22
Chebyshev Distance: 7
----------------------------------------------------------------------
Name:Sabu Bhuju
Lab No.:1
Roll No.535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe" "c:/Users/sabub/
Works/Lab2.py"
Original Nominal Dataset:
       Color     Shape Texture
ItemA    Red    Circle   Rough
ItemB   Blue    Square  Smooth
ItemC  Green  Triangle  Smooth
ItemD    Red    Circle   Rough

One-Hot Encoded Dataset:
       Color_Blue  Color_Green  Color_Red  Shape_Circle  Shape_Square  Shape_Triangle  Texture_Rough  Texture_Smooth
ItemA       False        False       True          True         False           False           True           False
ItemB        True        False      False         False          True           False          False            True
ItemC       False         True      False         False         False            True          False            True
ItemD       False        False       True          True         False           False           True           False

Cosine Similarity Matrix:
       ItemA     ItemB     ItemC  ItemD
ItemA    1.0  0.000000  0.000000    1.0
ItemB    0.0  1.000000  0.333333    0.0
ItemC    0.0  0.333333  1.000000    0.0
ItemD    1.0  0.000000  0.000000    1.0

Cosine Distance Matrix:
       ItemA     ItemB     ItemC  ItemD
ItemA    0.0  1.000000  1.000000    0.0
ItemB    1.0  0.000000  0.666667    1.0
ItemC    1.0  0.666667  0.000000    1.0
ItemD    0.0  1.000000  1.000000    0.0
----------------------------------------------------------------------
Name: Sabu Bhuju
Lab No.: 2
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe"
Works/Lab4.py"
1. Staff Personal Data:
   Staff_ID    Name  Age Department    JoinDate
0       101    Sabu   29         IT  2018-05-12
1       102     Emi   34         HR  2021-07-19
2       103     Ian   27      Sales  2022-01-10
3       104  Smriti   22    Finance  2017-09-25
4       105   Kripa   31         IT  2020-03-14

2. Staff Salary Data:
   Staff_ID  Salary  Bonus  Rating
0       101   75000   5000       4
1       102   82000   7000       5
2       104   90000   8500       4
3       106   88000   8000       3

3. Integrated Dataset (Inner Join):
   Staff_ID    Name  Age Department    JoinDate  Salary  Bonus  Rating
0       101    Sabu   29         IT  2018-05-12   75000   5000       4
1       102     Emi   34         HR  2021-07-19   82000   7000       5
2       104  Smriti   22    Finance  2017-09-25   90000   8500       4

----------------------------------------------------------------------
Name: Sabu Bhuju
Lab No.: 4
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe"
Works/Lab3.py"
Original Data:

     Name   Age Gender       City
0    Sabu  20.0      F  Kathmandu
1   Kripa   NaN      F    Pokhara
2     Emi  18.0      F    Pokhara
3  Smriti  22.0      F   Lalitpur
4    Sabu  20.0      F  Kathmandu
5   Kripa  21.0      F    Pokhara

After Handling Missing Values (Mean for Age):

     Name   Age Gender       City
0    Sabu  20.0      F  Kathmandu
1   Kripa  20.2      F    Pokhara
2     Emi  18.0      F    Pokhara
3  Smriti  22.0      F   Lalitpur
4    Sabu  20.0      F  Kathmandu
5   Kripa  21.0      F    Pokhara
```

```
After Removing Duplicates:

     Name   Age Gender       City
0    Sabu  20.0      F  Kathmandu
1   Kripa  20.2      F    Pokhara
2     Emi  18.0      F    Pokhara
3  Smriti  22.0      F   Lalitpur
5   Kripa  21.0      F    Pokhara

After Correcting Inconsistent Data:

     Name   Age Gender       City
0    Sabu  20.0      F  Kathmandu
1   Kripa  20.2      F    Pokhara
2     Emi  18.0      F    Pokhara
3  Smriti  22.0      F   Lalitpur
5   Kripa  21.0      F    Pokhara

----------------------------------------------------------------------
Name: Sabu Bhuju
Lab No.: 3
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.e
Works/Lab9.py"
Sample Dataset:
     Outlook Temperature Humidity  Windy PlayTennis
0      Sunny         Hot     High  False         No
1      Sunny         Hot     High   True         No
2   Overcast         Hot     High  False        Yes
3       Rain        Mild     High  False        Yes
4       Rain        Cool   Normal  False        Yes
5       Rain        Cool   Normal   True         No
6   Overcast        Cool   Normal   True        Yes
7      Sunny        Mild     High  False         No
8      Sunny        Cool   Normal  False        Yes
9       Rain        Mild   Normal  False        Yes
10     Sunny        Mild   Normal   True        Yes
11  Overcast        Mild     High   True        Yes
12  Overcast         Hot   Normal  False        Yes
13      Rain        Mild     High   True         No

Information Gain Calculation:
Entropy of the entire dataset (H(S)): 0.9403

Information Gain for 'Outlook' attribute: 0.2467
Information Gain for 'Temperature' attribute: 0.0292
Information Gain for 'Humidity' attribute: 0.1518
Information Gain for 'Windy' attribute: 0.0481

-------------------------------------------------
Name: Sabu Bhuju
Lab No.: 9
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe"
Works/Lab5.py"
1. Original Dataset
   Student_ID Program       City  Marks  Attendance
0         101     BIT  Kathmandu     72          82
1         102     BBA   Lalitpur     85          91
2         103     BIT  Bhaktapur     68          76
3         104     BBS  Kathmandu     90          95
4         105     BBA   Lalitpur     78          88
5         106     BIT  Bhaktapur     88          92

2a. Contingency Matrix (Program vs City)
City     Bhaktapur  Kathmandu  Lalitpur
Program
BBA              0          0         2
BBS              0          1         0
BIT              2          1         0

2b. After Min-Max Transformation on Marks
   Student_ID  Marks  Marks_MinMax
0         101     72      0.181818
1         102     85      0.772727
2         103     68      0.000000
3         104     90      1.000000
4         105     78      0.454545
5         106     88      0.909091
```

```
2c. After Z-Score Normalization on Attendance
   Student_ID  Attendance  Attendance_ZScore
0         101          82          -0.824042
1         102          91           0.566529
2         103          76          -1.751089
3         104          95           1.184560
4         105          88           0.103005
5         106          92           0.721037
----------------------------------------------------------------------
Name:Sabu Bhuju
Lab No.:5
Roll No.535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe"
Works/Lab7.py"

Frequent Itemsets:

L1:
{'Butter'} Support Count = 4
{'Diaper'} Support Count = 5
{'Milk'} Support Count = 7
{'Bread'} Support Count = 7
{'Beer'} Support Count = 5

L2:
{'Butter', 'Milk'} Support Count = 3
{'Butter', 'Bread'} Support Count = 3
{'Diaper', 'Milk'} Support Count = 3
{'Diaper', 'Beer'} Support Count = 4
{'Milk', 'Bread'} Support Count = 5
{'Milk', 'Beer'} Support Count = 3
{'Beer', 'Bread'} Support Count = 3
```

```
Strong Association Rules:

{'Butter'} -> {'Milk'} Confidence = 0.75
{'Butter'} -> {'Bread'} Confidence = 0.75
{'Diaper'} -> {'Milk'} Confidence = 0.6
{'Diaper'} -> {'Beer'} Confidence = 0.8
{'Beer'} -> {'Diaper'} Confidence = 0.8
{'Milk'} -> {'Bread'} Confidence = 0.71
{'Bread'} -> {'Milk'} Confidence = 0.71
{'Beer'} -> {'Milk'} Confidence = 0.6
{'Beer'} -> {'Bread'} Confidence = 0.6

--------------------------------------------------
Name: Sabu Bhuju
Lab No.: 7
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe
Works/Lab6.py"
1. Base Data (BTS Album Sales):
 Year      Region           Album  Sales_Thousands
 2018 South Korea   Love Yourself             2200
 2018         USA   Love Yourself             1800
 2019 South Korea Map of the Soul             2500
 2019         USA Map of the Soul             2000
 2020 South Korea              BE             3000
 2020         USA              BE             2400
 2021 South Korea              BE             3100
 2021         USA              BE             2700

2. Multi-Index Data Cube:
                                  Sales_Thousands
Year Region      Album
2018 South Korea Love Yourself               2200
     USA         Love Yourself               1800
2019 South Korea Map of the Soul             2500
     USA         Map of the Soul             2000
2020 South Korea BE                          3000
     USA         BE                          2400
2021 South Korea BE                          3100
     USA         BE                          2700

3. OLAP Roll-up (Total Sales per Year):
      Sales_Thousands
Year
2018             4000
2019             4500
2020             5400
2021             5800

4. OLAP Drill-down (Sales in 2020):
                   Sales_Thousands
Region      Album
South Korea BE                3000
USA         BE                2400

5. OLAP Slice (Sales for Album = 'BE'):
                  Sales_Thousands
Year Region
2020 South Korea             3000
     USA                     2400
2021 South Korea             3100
     USA                     2700

6. OLAP Dice (Region = 'USA' AND Album = 'BE'):
                   Sales_Thousands
Year Region Album
2020 USA    BE                2400
2021 USA    BE                2700

7. Pivot Table (Sales per Album per Year):
Album      BE  Love Yourself  Map of the Soul
Year
2018      NaN         4000.0              NaN
2019      NaN            NaN           4500.0
2020   5400.0            NaN              NaN
2021   5800.0            NaN              NaN

----------------------------------------------------------------------
Name: Sabu Bhuju
Lab No.: 6
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe"
Works/Lab8.py"
1. One-Hot Encoded Transactions:
    Beer  Bread  Butter   Coke  Diaper   Milk
0  False   True    True  False   False   True
1   True   True   False  False    True  False
2   True  False   False   True    True   True
3   True   True   False  False    True   True
4  False   True   False   True   False   True
5   True   True   False  False   False   True
6  False   True    True  False   False  False
7  False  False    True  False    True   True
8  False   True    True  False   False   True
9   True  False   False  False    True  False

2. Frequent Itemsets (min_support=0.3):
    support         itemsets
0       0.7           (Milk)
1       0.7          (Bread)
2       0.4         (Butter)
3       0.5         (Diaper)
4       0.5           (Beer)
5       0.5    (Milk, Bread)
6       0.3  (Butter, Bread)
7       0.3   (Milk, Butter)
8       0.3   (Milk, Diaper)
9       0.4   (Beer, Diaper)
10      0.3    (Beer, Bread)
11      0.3     (Beer, Milk)
```

```
3. Strong Association Rules (min_confidence=0.6):
{'Milk'} -> {'Bread'} | Confidence = 0.71
{'Bread'} -> {'Milk'} | Confidence = 0.71
{'Butter'} -> {'Bread'} | Confidence = 0.75
{'Butter'} -> {'Milk'} | Confidence = 0.75
{'Diaper'} -> {'Milk'} | Confidence = 0.6
{'Beer'} -> {'Diaper'} | Confidence = 0.8
{'Diaper'} -> {'Beer'} | Confidence = 0.8
{'Beer'} -> {'Bread'} | Confidence = 0.6
{'Beer'} -> {'Milk'} | Confidence = 0.6

----------------------------------------------------------------------
Name: Sabu Bhuju
Lab No.: 8
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe"
Works/Lab11.py"
1. Given Confusion Matrix:
[[ 50  25]
 [ 10 100]]

True Negatives (TN): 50
False Positives (FP): 25
False Negatives (FN): 10
True Positives (TP): 100

2. Calculated Performance Metrics:
Accuracy: 0.8108 or 81.08%
Precision: 0.8000
Recall (Sensitivity): 0.9091

-------------------------------------------------
Name: Sabu Bhuju
Lab No.: 11
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

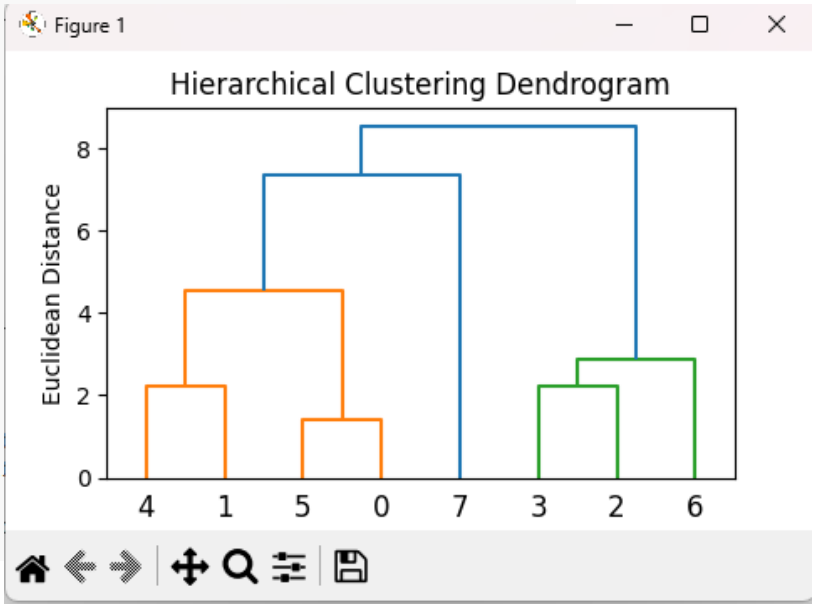
```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe" "
Works/Lab10.py"
1. Original Dataset:
  Weather Temperature Humidity    Wind PlayGame
0  Cloudy        Mild   Normal  Strong       No
1   Sunny        Warm     High    Weak      Yes
2   Rainy        Cold   Normal    Weak      Yes
3   Sunny        Cold     High  Strong       No
4  Cloudy        Warm     High    Weak      Yes
5   Rainy        Mild   Normal  Strong      Yes
6   Sunny        Warm   Normal    Weak      Yes
7  Cloudy        Mild     High    Weak       No
8   Rainy        Cold     High  Strong      Yes
9   Sunny        Warm   Normal    Weak      Yes

2. Encoded Dataset:
   Weather  Temperature  Humidity  Wind  PlayGame
0        0            1         1     0         0
1        2            2         0     1         1
2        1            0         1     1         1
3        2            0         0     0         0
4        0            2         0     1         1
5        1            1         1     0         1
6        2            2         1     1         1
7        0            1         0     1         0
8        1            0         0     0         1
9        2            2         1     1         1
```

```
3. Naive Bayes Classification Results:
Test Data:
   Weather  Temperature  Humidity  Wind
8        1            0         0     0
1        2            2         0     1
5        1            1         1     0
Actual Labels: [1 1 1]
Predicted Labels: [0 1 0]
Accuracy of the model: 0.33

Prediction for new instance [[2, 1, 0, 1]]: 'No'

--------------------------------------------------
Name: Sabu Bhuju
Lab No.: 10
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe" "c:/Users/s
Works/Lab12.py"

===== K-Means Clustering Result =====
   Age  Income  KMeans_Cluster
0   23   20000               1
1   25   22000               1
2   29   25000               1
3   35   35000               1
4   45   45000               0
5   52   52000               0
6   46   48000               0
7   48   50000               0
8   55   55000               0
9   60   60000               0

===== K-Median Clustering Result =====
   Age  Income  KMeans_Cluster  KMedian_Cluster
0   23   20000               1                1
1   25   22000               1                1
2   29   25000               1                1
3   35   35000               1                1
4   45   45000               0                0
5   52   52000               0                0
6   46   48000               0                0
7   48   50000               0                0
8   55   55000               0                0
9   60   60000               0                0
```

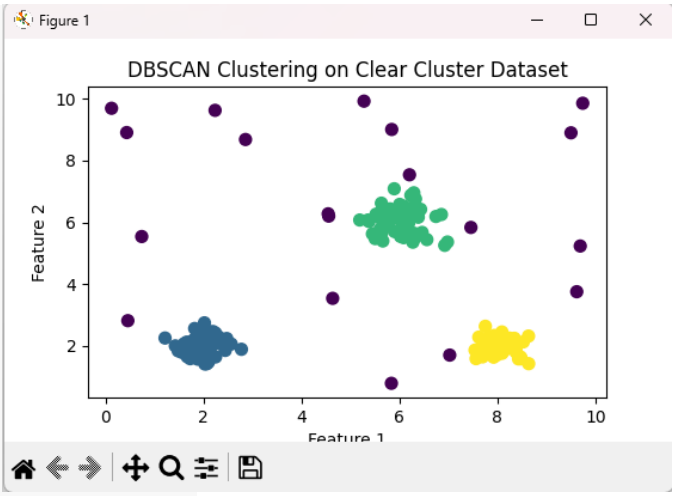
```
===== K-Mode Clustering Result =====
   Gender Profession       City  KMode_Cluster
0    Male    Student  Kathmandu              0
1  Female    Teacher   Lalitpur              1
2  Female    Teacher  Bhaktapur              1
3    Male   Engineer  Kathmandu              0
4    Male    Student   Lalitpur              0
5  Female   Engineer  Bhaktapur              1

--------------------------------------------------
Name: Sabu Bhuju
Lab No.: 12
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```

```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe"
Works/Lab13.py"
1. Sample Dataset for Hierarchical Clustering:
   X   Y
0  5  10
1  3   8
2  8   7
3  9   5
4  4   6
5  6   9
6  7   4
7  2   3

2. Cluster Assignments for 3 clusters:
   X   Y  Cluster
0  5  10        2
1  3   8        2
2  8   7        1
3  9   5        1
4  4   6        2
5  6   9        2
6  7   4        1
7  2   3        3

--------------------------------------------
Name: Sabu Bhuju
Lab No.: 13
Roll No.: 535/078
```



```
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks> & "C:\Program Files\Python313\python.exe" "c:/Users/sab
Works/Lab15.py"
1. Sample Dataset:
   Feature1  Feature2
0  5.496714  4.861736
1  5.647689  6.523030
2  4.765847  4.765863
3  6.579213  5.767435
4  4.530526  5.542560
5  4.536582  4.534270
6  5.241962  3.086720
7  3.275082  4.437712
8  3.987169  5.314247
9  4.091976  3.587696

2. Outlier Detection Results (first 10 rows):
   Feature1  Feature2  Outlier_Score  Outlier
0  5.496714  4.861736       0.316195    False
1  5.647689  6.523030       0.367617    False
2  4.765847  4.765863       0.190207    False
3  6.579213  5.767435       0.666844    False
4  4.530526  5.542560       0.206978    False
5  4.536582  4.534270       0.220743    False
6  5.241962  3.086720       0.184939    False
7  3.275082  4.437712       0.590634    False
8  3.987169  5.314247       0.289408    False
9  4.091976  3.587696       0.524984    False

--------------------------------------------------
Name: Sabu Bhuju
Lab No.: 15
Roll No.: 535/078
```

```
1. DBSCAN Clustering Results (first 10 rows):
   Feature1  Feature2  Original_Class  DBSCAN_Cluster
0  2.149014  1.958521               0               0
1  2.194307  2.456909               0               0
2  1.929754  1.929759               0               0
3  2.473764  2.230230               0               0
4  1.859158  2.162768               0               0
5  1.860975  1.860281               0               0
6  2.072589  1.426016               0               0
7  1.482525  1.831314               0               0
8  1.696151  2.094274               0               0
9  1.727593  1.576309               0               0

2. DBSCAN Plot Generated:
Please see the plot window for the visualization.

--------------------------------------------------
Name: Sabu Bhuju
Lab No.: 14
Roll No.: 535/078
PS C:\Users\sabub\OneDrive\Desktop\Sabu\8th sem\DWDM\LabWorks>
```



TRIBHUVAN UNIVERSITY

Padmakanya Multiple Campus

Bagbazar, Kathmandu

LAB REPORT ON
DATA WAREHOUSING
AND
DATA MININIG (BIT454)

Submitted By:
Name: Sabu Bhuju
Roll NO.: 535/078
Bit 8th semester

Submitted To:
Mr. Anil Lamichhane

CONTENTS

1. Write a program to compute Euclidean, Manhattan, and Chebyshev distances between two points.
2. Write a program to generate a distance matrix of nominal attributes using cosine similarity.
3. Write a program to perform data cleaning.
   a. Handle missing values using the mean
   b. Remove duplicates
   c. Correct the inconsistent data
4. Write a program to compute Write a program to perform data integration. Illustrate how you can combine two lists based on a common ID
5. Write a program to perform data transformation. Illustrate the following:
   a. Convert nominal data to a contingency matrix.
   b. Perform minimax transformation of data.
   c. Perform z-score normalization of data.
6. Write a program to illustrate OLAP operations.
7. Write a program to generate frequent item sets and strong association rules using the Apriori algorithm.
8. Write a program to generate frequent itemsets and strong association rules using the FP tree.
9. Write a program to compute info gain of an attribute in the given dataset.
10. Write a program to perform classification using a Naïve Bayesian classifier.
11. Write a program to compute accuracy, precision, and recall from the given confusion matrix.
12. Write a program to implement partitioning methods. Illustrate following
    a. K-means algorithm
    b. K-median algorithm
    c. K-mode algorithm
13. Write a program to illustrate hierarchical clustering.
14. Write a program to illustrate DBSCAN algorithm.
15. Write a program to illustrate proximity-based outlier detection.